SANJANA R

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PROFESSIONAL SUMMARY

MCA student specializing in Artificial Intelligence and Generative AI with hands-on experience in Python, Java, C++, JavaScript, SQL, React, REST APIs, and backend development. Experienced in building full-stack and AI-powered applications using FastAPI, React, databases, and third-party APIs, with strong foundations in data structures, problem solving, debugging, and software development.

EDUCATION

Master of Computer Applications (MCA) – Artificial Intelligence and Generative AI 2025 – 2027
Alliance University, Bengaluru

Bachelor of Computer Applications (BCA) 2022 – 2025
Seshadripuram Degree College, University of Mysore — CGPA: 9.21

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL, JavaScript
Frontend: React.js, HTML5, CSS3, JavaScript, Vite
Backend: FastAPI, REST APIs, Node.js
Databases: MySQL, SQL, ChromaDB
ML/DL: Scikit-Learn, TensorFlow, PyTorch, CNNs, Transformers, NLTK, Ensemble Learning,YOLO
Tools: Git, AWS,GitHub, Streamlit, Hugging Face, Vercel, Figma

EXPERIENCE

AI Intern – TechSaksham (Microsoft & SAP CSR Initiative) Jan 2025 – Apr 2025
Edunet Foundation — AICTE

- Completed the AI: Transformative Learning with TechSaksham internship program.
- Gained foundational knowledge of artificial intelligence concepts and their practical applications through structured learning activities.

PROJECTS

FashionVerse – Adaptive AI Fashion Stylist 2026
PPO, API, Generative AI, 3D Virtual Try-On

- Engineered an adaptive AI fashion stylist generating personalized outfit recommendations using user preferences, context, and interaction feedback.
- Developed a Proximal Policy Optimization (PPO) reinforcement learning agent to adapt recommendations based on user feedback.
- Integrated Gemini for generative styling insights and natural-language recommendations with 3D virtual try-on visualization.

NutriLens AI – Intelligent Nutrition Assistant 2026
Python, FastAPI, REST APIs, ChromaDB, Gemini, JWT, Pydantic

- Built a backend-driven AI application using FastAPI and REST APIs to process product information and generate personalized nutrition insights.
- Integrated external product data with ChromaDB semantic retrieval and Gemini to build a retrieval-augmented application..
- Implemented JWT authentication and Pydantic validation for secure API requests and structured data handling..

AI-Based Surgical Instrument Detection and Segmentation 2026
Python, PyTorch, U-Net, YOLO, ROS 2

- Developed a computer vision pipeline using U-Net and YOLO for surgical instrument segmentation and detection on the EndoVis2017 robotic surgery dataset.
- Achieved 93.17% Dice score and 87.22% IoU for instrument segmentation, with 92.43% precision and 93.93% recall.
- Integrated the vision pipeline with ROS 2 and Gazebo, using a simulated UR10e robotic arm for robotic surgery experimentation and control.

LEADERSHIP ACTIVITIES

Red Cross President 2023 – 2024

- Led 20+ community initiatives, including blood donation drives and social outreach programs.

Treasurer, Rotaract Club 2022 – 2025

- Organized and supported social impact and community engagement events.

ACHIEVEMENTS

- Winner – Inter-College Hackathon
- Winner – Multiple Inter-College Coding Competitions